Nama: nismara ayudia

Npm: 5520125056

Kelas: IF B 2025

```
package main.java;
import java.util.Scanner;

public class MainJava {

  // Method untuk menghitung nilai akhir
  public static double hitungNilaiAkhir(double tugas, double uts, double uas) {
    return (0.30 * tugas) + (0.30 * uts) + (0.40 * uas);
  }

  public static void main(String[] args) {

    Scanner input = new Scanner(System.in);

    System.out.println("================================================");
    System.out.println("      SISTEM PENGOLAHAN NILAI MAHASISWA");
    System.out.println("================================================");

    // Input jumlah mahasiswa
    System.out.print("Masukkan jumlah mahasiswa : ");
    int jumlah = input.nextInt();
    input.nextLine();

    // Array untuk menyimpan data mahasiswa
    String[] npm = new String[jumlah];
    String[] nama = new String[jumlah];

    double[] tugas = new double[jumlah];
    double[] uts = new double[jumlah];
    double[] uas = new double[jumlah];
    double[] nilaiAkhir = new double[jumlah];
```

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char[] huruf = new char[jumlah];
String[] status = new String[jumlah];

// Loop input data mahasiswa
for (int i = 0; i < jumlah; i++) {

    System.out.println();
    System.out.println("Data Mahasiswa Ke-" + (i + 1));
    System.out.println("--------------------------------");

    System.out.print("NPM        : ");
    npm[i] = input.nextLine();

    System.out.print("Nama       : ");
    nama[i] = input.nextLine();

    // Input nilai Tugas dengan validasi
    tugas[i] = inputNilai(input, "Nilai Tugas");

    // Input nilai UTS dengan validasi
    uts[i] = inputNilai(input, "Nilai UTS");

    // Input nilai UAS dengan validasi
    uas[i] = inputNilai(input, "Nilai UAS");

    // Hitung nilai akhir
    nilaiAkhir[i] = hitungNilaiAkhir(tugas[i], uts[i], uas[i]);

    // Menentukan nilai huruf
    if (nilaiAkhir[i] >= 80) {
        huruf[i] = 'A';
    } else if (nilaiAkhir[i] >= 70) {
        huruf[i] = 'B';
    } else if (nilaiAkhir[i] >= 60) {
        huruf[i] = 'C';
    } else if (nilaiAkhir[i] >= 50) {
```

```
            huruf[i] = 'D';
        } else {
            huruf[i] = 'E';
        }

        // Menentukan status kelulusan
        if (huruf[i] == 'A' || huruf[i] == 'B' || huruf[i] == 'C') {
            status[i] = "LULUS";
        } else {
            status[i] = "TIDAK LULUS";
        }
    }

    // Menampilkan hasil
    System.out.println();
    System.out.println("=================================================================");
    System.out.println("                DAFTAR NILAI MAHASISWA");
    System.out.println("=================================================================");

    System.out.printf("%-4s %-12s %-15s %7s %7s %7s %8s %7s %-12s%n",
            "No", "NPM", "Nama", "Tugas", "UTS", "UAS",
            "Akhir", "Huruf", "Status");

    System.out.println("-----------------------------------------------------------------");

    for (int i = 0; i < jumlah; i++) {

        System.out.printf("%-4d %-12s %-15s %7.0f %7.0f %7.0f %8.1f %7c %-12s%n",
                (i + 1),
                npm[i],
                nama[i],
                tugas[i],
                uts[i],
                uas[i],
                nilaiAkhir[i],
                huruf[i],
```

```
            status[i]);
    }

    System.out.println("================================================================");

    // Statistik nilai
    int jumlahLulus = 0;
    int jumlahTidakLulus = 0;

    double nilaiTertinggi = nilaiAkhir[0];
    double nilaiTerendah = nilaiAkhir[0];
    double totalNilai = 0;

    for (int i = 0; i < jumlah; i++) {

        totalNilai += nilaiAkhir[i];

        if (nilaiAkhir[i] > nilaiTertinggi) {
            nilaiTertinggi = nilaiAkhir[i];
        }

        if (nilaiAkhir[i] < nilaiTerendah) {
            nilaiTerendah = nilaiAkhir[i];
        }

        if (status[i].equals("LULUS")) {
            jumlahLulus++;
        } else {
            jumlahTidakLulus++;
        }
    }

    double rataRata = totalNilai / jumlah;

    System.out.println();
    System.out.println("STATISTIK NILAI");
```

```
    System.out.println("---------------------------------");
    System.out.println("Jumlah Mahasiswa       : " + jumlah);
    System.out.println("Jumlah Mahasiswa Lulus : " + jumlahLulus);
    System.out.println("Jumlah Tidak Lulus     : " + jumlahTidakLulus);
    System.out.printf("Nilai Tertinggi        : %.1f%n", nilaiTertinggi);
    System.out.printf("Nilai Terendah         : %.1f%n", nilaiTerendah);
    System.out.printf("Rata-rata Kelas        : %.2f%n", rataRata);

    System.out.println("================================================================");
    System.out.println("              PROGRAM SELESAI");
    System.out.println("================================================================");

    input.close();
}

// Method untuk input dan validasi nilai
public static double inputNilai(Scanner input, String jenisNilai) {

    double nilai;

    while (true) {

        System.out.print(jenisNilai + " : ");
        nilai = input.nextDouble();

        if (nilai >= 0 && nilai <= 100) {
            break;
        } else {
            System.out.println("Nilai tidak valid!");
            System.out.println("Nilai harus berada pada rentang 0 - 100.");
            System.out.println("Silakan masukkan nilai kembali.");
        }
    }

    input.nextLine();
```

```
    return nilai;
  }
}
```

```
run:
==================================================
       SISTEM PENGOLAHAN NILAI MAHASISWA
==================================================
Masukkan jumlah mahasiswa : 4

Data Mahasiswa Ke-1
---------------------------------
NPM          : 5520125056
Nama         : nismara ayudia
Nilai Tugas : 90
Nilai UTS : 85
Nilai UAS : 90

Data Mahasiswa Ke-2
---------------------------------
NPM          : 5520125059
Nama         : rani maelani
Nilai Tugas : 80
Nilai UTS : 90
Nilai UAS : 85

Data Mahasiswa Ke-3
---------------------------------
NPM          : 5520125051
Nama         : anggi
Nilai Tugas : 85
Nilai UTS : 95
Nilai UAS : 85
```

```
Data Mahasiswa Ke-4
--------------------------------
NPM          : 5520125041
Nama         : arvi
Nilai Tugas : 90
Nilai UTS : 90
Nilai UAS : 85

=====================================================================
                      DAFTAR NILAI MAHASISWA
=====================================================================
No   NPM          Nama               Tugas     UTS     UAS    Akhir   Huruf Status
---------------------------------------------------------------------------
1    5520125056   nismara ayudia       90      85      90     88,5       A LULUS
2    5520125059   rani maelani         80      90      85     85,0       A LULUS
3    5520125051   anggi                85      95      85     88,0       A LULUS
4    5520125041   arvi                 90      90      85     88,0       A LULUS
=====================================================================

STATISTIK NILAI
--------------------------------
Jumlah Mahasiswa       : 4
Jumlah Mahasiswa Lulus : 4
Jumlah Tidak Lulus     : 0
Nilai Tertinggi        : 88,5
Nilai Terendah         : 85,0
Rata-rata Kelas        : 87,38
=====================================================================
              PROGRAM SELESAI
=====================================================================
```